

Integration Vulnerabilities — Comprehensive Test Plan

Infrastructure & Security Upgrade Validation

Jira Epic	
Branch / Build	
Impacted Areas	Events API, CTMS Study Mapping
Services in Scope	5 Deployable Integration Services
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Environment Coverage	Feature Branch · Stage · Production

ENVIRONMENT LEGEND

FB Full regression + negative + break tests	Master Smoke + sanity + backward compatibility	Stage Happy-path + smoke only — no data writes
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1. Objective & What Changed

This test plan validates that the infrastructure and security upgrade (EDC-42647) does NOT introduce any regression across the 5 deployable integration services and all EDC consumer touchpoints. There is zero new business logic — every failed test indicates a regression caused by the upgrade.

1.1 Changes Under Test

Change Category	From → To	QA Risk / Why It Matters
Runtime	.NET 6 → .NET 10	Runtime behavior differences: HTTP client timeouts, serialization, thread-pool defaults. Any 5xx that did not exist before is a regression.
Auth Library	IdentityServer4.AccessTokenValidation → IdentityModel.AspNetCore.OAuth2Introspection 6.2.0	Token validation path has completely changed. A misconfigured introspection endpoint causes all EDC→Integrations calls to silently fail (401/403 or no response).
Backchannel Header	N/A → MedriOdpCustomHeaderDelegatingHandler adds Medrio-UserType header; 5s timeout	If header is missing or malformed, IDP may reject introspection. If timeout is wrong, long IDP latency causes cascading failures.
ORM / DB Driver	EF Core (prev) → EF Core / Microsoft.Data.SqlClient (new ver)	SQL query generation can differ. Watch for: query timeouts, missing eager-loads, implicit transactions changed.
Message Bus	NServiceBus (prev ver) → NServiceBus (new ver)	Queue message serialization format may differ. Existing paused/queued messages must still be processable.
Logging	Serilog (prev ver) → Serilog (new ver)	Log levels or sink configs may differ — affects observability but also could mask or expose PII.
API Docs	Swagger root redirect existed → root→swagger redirect REMOVED	Swagger must still open via /swagger/index.html. Root (/) must NOT redirect to swagger (security).
Docker base image	.NET 6 base → .NET 10 base	OS-level differences may cause cert-store, timezone, or file-permission issues.

1.2 Services Under Test

#	Service Name	Purpose	Test Sections
1	Integrations Public API	Handles CTMS redirect (/api/v1/redirect/*)	TC-REDIR-*
2	Integrations Internal Management API	Subscription CRUD (/api/v1/study/{id}/subscriptions/*), CTMS mapping (/api/v1/mapping/*)	TC-SUBS-*, TC-CTMS-*
3	Integrations Events API	Spool/Status/Sync endpoints (/Subscription/Spool, /Synchronization/*)	TC-EVTAPI-*
4	Events.Service (NServiceBus handler)	Processes ISubscriptionEvent messages from queue; delivers callbacks to subscribers	TC-EVT-*, TC-RETRY-*
5	PausedSubscriptionsRetryProcessor	Background job: drains paused queue, triggers status transitions, sends emails	TC-RETRY-*

2. Test Data Setup (Mandatory Before Execution)

■ **NOTE:** Complete ALL setup steps before running any test case. Skipping setup will cause false failures.

2.1 Feature Flag

Enable the EventsApi feature flag for the test tenant:

- Flag: EventsApi = 20140a70-c0de-40bc-b4c2-86cc4f0456ea (Sample GUID)
- Scope: Enabled for the primary test tenant / study used in FB/Stage
- Verification: Admin Portal → Subscription Configuration tab must appear for test studies

2.2 Studies Required

Study	Configuration	Use In Test Cases	Critical Setup Check
Study A	CTMS: ON, Mapping: SET, Events: ON, ≥1 ACTIVE subscription pointing to WORKING endpoint	TC-SUBS-*, TC-CTMS-001,002, TC-REDIR-001, TC-EVT-*, TC-BACK-001	Load /api/v1/study/{id}/subscriptions → 200 with ≥1 subscription returned
Study B	Events: ON, Subscriptions: NONE (empty list)	TC-SUBS-005 (negative), TC-EVT-005 (no-delivery negative)	Load /api/v1/study/{id}/subscriptions → 200 with empty array []
Study C	CTMS: ON, Mapping: NOT SET (empty), Events: OFF	TC-CTMS-004, TC-REDIR-002	Load study detail page → CTMS Customer ID field shows empty/blank, no error
Study D	CTMS: OFF, Events: OFF (plain study)	TC-BACK-002 (no integrations calls expected)	Load study detail → NO CTMS section visible, no integrations API called

2.3 Subscriber Endpoints

- Working endpoint: An HTTP server that accepts POST callbacks and returns HTTP 200. Use a tool like a team-controlled test receiver. Record the URL.
- Failing endpoint: Any URL that returns a non-2xx (e.g., http://localhost:9999/dead) OR times out. Record the URL. This endpoint is used to trigger retry/paused/stopped flows.
- Before starting TC-EVT tests: Confirm the working endpoint is reachable from the integration service network (no firewall block).

2.4 Access & Tooling

- EDC Admin user: Full admin rights to Admin Portal, ability to trigger ODM/Excel/data entry imports and run Recalc jobs.
- Postman / HTTP client: Configured with a valid OAuth2 token and an expired/invalid token. Both are needed for TC-AUTH negative tests.
- RabbitMQ management console: Access to view the paused queue during TC-RETRY tests.
- Email/log access: Test mailbox or log viewer to verify notification emails in TC-RETRY-004 and TC-EVTAPI-002.
- Logs access: Read access to integration service logs (structured JSON) for all 5 services.

3. Test Cases — Execution Checklist

■ **NOTE:** Each test case row contains: TC ID + name, environment applicability (QA / STG / PROD), priority, numbered steps, and exact pass criteria. Mark Status as: ☒ Pass | ☒ Fail | ☒ Skip (with reason) | ☒ Blocked.

3.1 Authentication — OAuth2 Introspection Regression (TC-AUTH-*)

Risk area: The entire auth library was replaced. A broken introspection config causes ALL EDC→Integrations calls to silently fail. Test auth FIRST before any other section.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-AUTH-001 <i>Valid token – end-to-end OAuth2 introspection</i>	P1-Critical QA · STG · PROD Status: ☐ Open	- Pre-condition: Obtain a VALID active OAuth2 access token (via normal EDC login). - Open Admin Portal → navigate to Study A → Subscription Configuration tab. - Open browser DevTools → Network tab. Load the subscription list. - Inspect the outbound call to /api/integrations/study/{id}/subscriptions → confirm request completes. - In integration service logs, confirm: token introspection call was made to IDP, response was active=true, and Medrio-UserType header was present in the backchannel call. - Repeat for: CTMS mapping load (Study A detail page) and Go to CTMS redirect (Study A).	- HTTP 200 returned for all three EDC→Integrations calls. No 401 or 403. - Logs show introspection succeeded (active=true) for each call. - Logs confirm Medrio-UserType header was sent in the IDP backchannel request. - No 500 errors anywhere in the call chain.
TC-AUTH-002 <i>Expired/invalid token – proper 401 returned, no data leak</i>	P1-Critical QA Status: ☐ Open	- Pre-condition: Obtain an EXPIRED or deliberately invalid Bearer token (tamper the last 10 chars in Postman). - Call POST /api/v1/study/{studyId}/subscriptions directly via Postman with Authorization: Bearer <invalid_token>. - Call GET /api/v1/mapping/{studyId} with the same invalid token. - Call GET /api/v1/redirect/integration with the same invalid token. - In each case, inspect the HTTP response body and headers carefully.	- HTTP 401 returned for all three calls. HTTP 403 is also acceptable (auth rejection). HTTP 500 is a DEFECT. - Response body contains NO stack trace, NO internal service details, NO SQL fragments, NO exception message. - Logs show introspection returned active=false and request was rejected cleanly. - No IDP introspection error propagated to the HTTP response (fail closed).
TC-AUTH-003 <i>IDP introspection timeout – service fails closed within 5 seconds</i>	P2-High QA Status: ☐ Open	- Pre-condition: Simulate IDP introspection being slow (use a proxy like Charles/Toxiproxy to add 10s delay on the IDP endpoint) OR confirm this via reviewing the MedrioIdpCustomHeaderDelegatingHandler timeout config. - With the delay active, make a valid API call to GET /api/v1/study/{studyId}/subscriptions. - Observe the response time and HTTP status code.	- Request fails with HTTP 401 or 503 within ~5 seconds (backchannel timeout), NOT after 30+ seconds. - Service does NOT hang indefinitely — next request after timeout also completes (service is not deadlocked). - Logs show 'timeout' or equivalent against the introspection endpoint, not a generic unhandled exception. - No queue backup or service crash resulted from the timeout event.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
		- Check integration service logs for timeout behavior. - Remove the delay and confirm service recovers normally on the next call.	
TC-AUTH-004 <i>Medrio-UserType header injection – IDP backchannel verification</i>	P2-High QA · STG Status: ☐ Open	- Pre-condition: Enable verbose/debug logging on the integration service or set up a network trace on the IDP-facing interface. - From Admin Portal, perform a normal authenticated action (load subscriptions for Study A). - In logs or network trace, find the outbound HTTP call from the integration service to the IDP introspection endpoint. - Inspect the request headers of that outbound call.	- The outbound introspection request includes the header Medrio-UserType with a non-empty value. - IDP accepts the request (introspection returns active=true). - No auth regression — the overall EDC action completes successfully.

3.2 Subscription Configuration UI (TC-SUBS-*)

Risk area: CRUD operations through the Management API must survive the .NET 10 / EF Core upgrade without behavioral changes to persistence.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-SUBS-001 <i>View subscription list – Study A loads without errors</i>	P1-Critical QA · STG · PROD Status: ☐ Open	- Pre-condition: Study A configured (CTMS ON, ≥1 active subscription, EventsApi flag ON). - Log in as EDC admin → Admin Portal → select Study A → Subscription Configuration tab. - Observe: does the list load? Are existing subscriptions visible? - Open browser DevTools → Network → find the XHR to /api/integrations/study/{id}/subscriptions → confirm HTTP 200 and valid JSON array.	- Subscription list renders with all configured subscriptions listed. - HTTP 200 from backend. Response body is a valid JSON array (not empty, not null). - No UI console errors (JavaScript exceptions in browser DevTools). - Page load time under 5 seconds for normal dataset.
TC-SUBS-002 <i>Create new subscription – working endpoint, saved correctly</i>	P1-Critical QA · STG Status: ☐ Open	- Pre-condition: On Study A Subscription Configuration tab. - Click 'Add Subscription' (or equivalent UI button). - Enter: endpoint URL = WORKING endpoint URL, select at least 1 event type, save. - After save, refresh the page (F5 or navigate away and back). - Confirm the newly created subscription appears with correct endpoint URL and event type. - In integration service logs, confirm POST /api/v1/study/{id}/subscriptions returned 201.	- HTTP 201 from Management API on create. - Subscription persists after page refresh with correct URL and event type(s). - No duplicate subscription created (only 1 new entry visible). - No 500 or unhandled exception in logs.
TC-SUBS-003 <i>Edit subscription – update</i>	P2-High QA · STG Status: ☐ Open	- Pre-condition: At least one subscription exists for Study A (created in TC-SUBS-002 or pre-existing). - Open the existing subscription in edit mode → change the endpoint URL to a new valid URL.	- HTTP 200/204 from Management API on update. - New URL persisted; original URL replaced (not duplicated).

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
<i>endpoint URL, change persists</i>		- Save the changes. - Refresh the page and re-open the subscription. - Verify the new URL is shown, and the old URL is no longer present.	Existing event type selections are preserved (not reset by the save).
TC-SUBS-004 <i>Delete subscription – removed from list, not re-appearing</i>	P2-High QA · STG Status: ☐ Open	- Pre-condition: At least one subscription exists that is safe to delete. - Record the subscription ID (from DevTools or logs before deletion). - Click 'Delete' on the subscription → confirm the deletion prompt if shown. - After deletion, immediately check the list (no refresh yet) — subscription should be gone. - Refresh the page — subscription must still be absent. - Make a GET /api/v1/study/{id}/subscriptions via Postman — confirm the deleted subscription ID is NOT in the response.	- HTTP 200/204 from Management API on delete. - Subscription absent immediately and after page refresh. - Direct API call confirms the ID is fully removed from storage.
TC-SUBS-005 <i>View subscriptions – Study B (no subscriptions, empty state)</i>	P2-High QA Status: ☐ Open	- Open Subscription Configuration for Study B (Events ON, subscriptions NONE). - Observe the UI rendering with an empty subscription list. - Check DevTools → the XHR returns HTTP 200 with an empty array [].	- UI shows an empty state message (e.g., 'No subscriptions configured') — NOT a blank screen, NOT an error. - HTTP 200 with [] (not 404, not 500). - No JavaScript console errors thrown for empty list handling.
TC-SUBS-006 <i>Create subscription with failing endpoint – failure captured, retry triggered</i>	P1-Critical QA Status: ☐ Open	- Pre-condition: Study A exists. Failing endpoint URL is ready. - Create a new subscription pointing to the FAILING endpoint URL. - Trigger an event (perform an ODM import or form submission on Study A). - Wait 30–60 seconds for the event processing cycle. - In integration service logs, look for: delivery attempt, HTTP failure recorded, subscription status change. - Check RabbitMQ management console — confirm messages are in the paused queue for this subscription. - Check email mailbox — confirm a failure notification email is generated (see TC-EVTAPI-002 for detail).	- Delivery failure is captured in logs (not silently dropped). - Subscription status transitions from Active to Paused (verify via GET /api/v1/study/{id}/subscriptions). - Messages visible in RabbitMQ paused queue. - Failure notification email sent with correct recipient and subscription details.

3.3 CTMS Customer ID Mapping (TC-CTMS-*)

Risk area: Mapping persistence through EF Core upgrade. Any query generation change could silently return stale or empty data.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-CTMS-001 <i>CTMS mapping loads on Study Detail page</i>	P1-Critical QA · STG · PROD Status: ☐ Open	- Open Admin Portal → Study Detail page for Study A. - Locate the CTMS Customer ID field. - In DevTools → Network → find the call to /api/v1/mapping/{studyId} or equivalent proxy endpoint. - Confirm it returns HTTP 200 with a non-empty mapping value.	- CTMS Customer ID field displays the configured mapping value (not blank, not 'undefined'). - HTTP 200 from /api/v1/mapping/{studyId}. - Page load does not stall waiting for the mapping call (completes within normal page load time).
TC-CTMS-002 <i>Set/update CTMS Customer ID – persists after save</i>	P1-Critical QA · STG Status: ☐ Open	- Open Study A detail page → CTMS Customer ID field. - Note the current value. Enter a NEW value (e.g., append '-QA-TEST' to distinguish). - Save the study detail page. - Navigate away (open a different study) → return to Study A. - Verify the new CTMS Customer ID value is shown. - Via Postman GET /api/v1/mapping/{studyId}, confirm the response body contains the new value.	- HTTP 200/204 from mapping PUT/POST on save. - New value shown after navigation (confirming DB persistence, not just UI state). - Postman direct API call confirms the value is the one just saved.
TC-CTMS-003 <i>Clear CTMS Customer ID – returns to empty/default state</i>	P2-High QA Status: ☐ Open	- Open Study A detail page → CTMS Customer ID field (has a value set). - Clear the field (delete all text) and save. - Navigate away and return to Study A. - Check the CTMS Customer ID field and Postman GET /api/v1/mapping/{studyId}.	- Field is empty/blank after navigation — not showing the old value. - API returns empty or null mapping — no stale data returned. - Go to CTMS link/button behavior reflects unmapped state (see TC-REDIR-002).
TC-CTMS-004 <i>Study C (no mapping) – graceful empty state, no errors</i>	P2-High QA Status: ☐ Open	- Open Study C detail page (CTMS ON, mapping NOT set). - Observe the CTMS Customer ID field. - Check DevTools → Network call to /api/v1/mapping/{studyId}.	- CTMS Customer ID field shows empty/placeholder — NOT an error message, NOT a JavaScript exception. - API returns HTTP 200 with null/empty body (NOT 404, NOT 500). - No unhandled exception popup in the UI.

3.4 Go to CTMS Redirect (TC-REDIR-*)

Risk area: Public API redirect must authenticate correctly with the new OAuth2 introspection and produce valid redirect URLs.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-REDIR-001 <i>Go to CTMS redirect – Study A (mapped) succeeds</i>	P1-Critical QA · STG · PROD Status: ☐ Open	- Pre-condition: Study A has CTMS ON and mapping SET. - From Global Navigation (or Study A page), click 'Go to CTMS'. - Observe the full redirect flow — track the intermediate URLs via DevTools or browser address bar. - Confirm the final landing URL is the correct CTMS destination for Study A's customer ID.	- HTTP 302/301 redirect issued by /api/v1/redirect/integration with a valid Location URL. - Final URL matches the expected CTMS URL for Study A's mapped customer ID. - No auth error in the redirect chain (no 401 at redirect endpoint). - Browser lands on CTMS page (or expected endpoint) without error.
TC-REDIR-002 <i>Go to CTMS – Study C (no mapping) safe failure</i>	P1-Critical QA Status: ☐ Open	- Pre-condition: Study C has CTMS ON but NO mapping. - Attempt to click 'Go to CTMS' for Study C. - Observe the UI response — is the button disabled? Is there a UI message? - If the button is enabled and clickable: click it and observe the HTTP response code.	- Either: the button is disabled/hidden when no mapping exists (preferred UX), OR button is enabled but returns a user-friendly error message. - NO HTTP 500 from the redirect endpoint. - NO unhandled JavaScript exception in the browser. - NO redirect to an invalid or blank URL.
TC-REDIR-003 <i>Redirect endpoint security – no valid auth returns 401</i>	P1-Critical QA Status: ☐ Open	- In Postman, call GET /api/v1/redirect/integration (or /api/v1/redirect/edc) with NO Authorization header. - Then call with an expired/invalid token. - Inspect the response HTTP status, body, and headers for both cases.	- HTTP 401 returned for both unauthenticated calls. - No redirect issued without authentication — zero data or URL leakage. - Response body has NO internal details, paths, or customer data.

3.5 Event Generation — EDC Job Triggers (TC-EVT-*)

Risk area: EDC job outputs must still route through NServiceBus and be processed by Events.Service. The new NServiceBus version must be backward-compatible with existing message serialization.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-EVT-001 <i>ODM import → event generated and delivered</i>	P1-Critical QA · STG Status: ☐ Open	- Pre-condition: Study A has active subscription to WORKING endpoint. Confirm endpoint is receiving requests. - Trigger an ODM import on Study A (upload a valid ODM XML file via Admin Portal). - Wait up to 60 seconds for event processing. - Check WORKING endpoint receiver logs — was a POST callback received?	- WORKING endpoint receives an HTTP POST callback with event payload. - Events.Service logs show: message consumed from queue, delivery attempted, HTTP 200 from subscriber, message acknowledged.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
		5. Check Events.Service logs — confirm ISubscriptionEvent was picked up and processed.	No error or exception in the Events.Service logs for this delivery.
TC-EVT-002 <i>Excel import → event generated and delivered</i>	P1-Critical QA Status: ☐ Open	- Trigger an Excel import on Study A (upload a valid template Excel file). - Wait up to 60 seconds. - Check WORKING endpoint receiver and Events.Service logs for delivery confirmation.	- Same delivery confirmation as TC-EVT-001 — callback received, no errors. - Confirm the event payload identifies the event source as Excel import (not mislabeled as ODM).
TC-EVT-003 <i>Data entry / form submission → event generated and delivered</i>	P1-Critical QA Status: ☐ Open	- On Study A, open a form and submit valid data entry. - Wait up to 60 seconds. - Check WORKING endpoint and Events.Service logs.	- Callback delivered to WORKING endpoint. - Event payload correctly represents a form-submission event (subject, event type field correct).
TC-EVT-004 <i>RecalculateVariablesJob → event generated and delivered</i>	P2-High QA Status: ☐ Open	- Trigger RecalculateVariablesJob on Study A (via admin job runner or direct API trigger as feasible in env). - Wait up to 120 seconds (background jobs may have longer latency). - Check Events.Service logs and WORKING endpoint.	- Event generated and delivered to WORKING endpoint. - No delivery failure for a job-triggered event (confirms job-to-event pipeline still intact after runtime upgrade).
TC-EVT-005 <i>Event trigger on Study B (no subscriptions) – no delivery attempted</i>	P2-High QA Status: ☐ Open	- Trigger an ODM import on Study B (Events ON, subscriptions NONE). - Wait 60 seconds. - Check Events.Service logs — should show event received but zero delivery attempts. - Check WORKING endpoint — should receive NO new calls from Study B.	- Events.Service logs show: event processed, 0 subscribers found, no delivery attempted. - No error logged — zero-subscriber case handled gracefully. - WORKING endpoint receives no new POST for Study B's event.

3.6 Events API Smoke & Email Notifications (TC-EVTAPI-*)

Risk area: Events API endpoints must respond after the .NET 10 upgrade. Failed subscription emails are a key customer-facing feature — any regression here is customer-visible.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-EVTAPI-001 <i>Events API smoke – Spool, Status, Synchronization endpoints respond</i>	P1-Critical QA · STG Status: ☐ Open	- Using Postman with a VALID token, call: POST /Subscription/Spool (with minimal valid body), GET /Subscription/Status. - Then call: GET /Synchronization/Complete, GET /Synchronization/Status, POST /Synchronization/Fault (with minimal body). - For each call, record the HTTP status code and response body.	- Each endpoint returns an expected 2xx status (or the specific documented status for that endpoint — not 401, not 500). - No endpoint returns a stack trace or internal exception. - Auth passes for all calls with valid token (confirms OAuth2 introspection works for Events API service too).

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-EVTAPI-002 <i>Failed subscription → notification email sent with correct content</i>	P1-Critical QA Status: ☐ Open	1. Pre-condition: A subscription pointing to FAILING endpoint exists for Study A (from TC-SUBS-006 or create fresh). 2. Trigger a new event (ODM import on Study A). 3. Wait up to 3 minutes for the failure cycle to complete. 4. Open the test mailbox — check for a new email notification. 5. Inspect email: recipient address, subject line, body content. 6. Confirm subscription status in GET /api/v1/study/{id}/subscriptions has changed.	• Email is received in the test mailbox (not spam-filtered — check spam if missing). • Email subject matches the expected format established in the previous build (no subject-line regression). • Email body references the correct Study ID or subscription identifier — not a blank or generic message. • Subscription status changes from Active to Paused.

3.7 PausedSubscriptionsRetryProcessor (TC-RETRY-*)

Risk area: Background job must start, fire on schedule, drain the paused queue, execute status transitions, and send emails — all of which rely on the upgraded runtime and NServiceBus.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-RETRY-001 <i>PausedSubscriptionsRetryProcessor fires on schedule</i>	P1-Critical QA Status: ☐ Open	1. Pre-condition: A subscription is in PAUSED state (result of TC-SUBS-006 or TC-EVTAPI-002). 2. In integration service logs, look for PausedSubscriptionsRetryProcessor service starting up and cycle trigger log lines. 3. Confirm at least one cycle fires within the configured interval (check config for interval — typically every few minutes).	• Log line confirms retry processor woke up and scanned for paused subscriptions. • No crash or unhandled exception in the background service log. • Service continues running after the cycle (does not exit or crash loop).
TC-RETRY-002 <i>Paused queue drain — messages are reprocessed on retry</i>	P1-Critical QA Status: ☐ Open	1. Pre-condition: Paused messages exist in RabbitMQ (from TC-SUBS-006). Confirm via RabbitMQ console. 2. FIX the failing endpoint — change the subscription's endpoint URL to the WORKING endpoint (or temporarily make the failing endpoint return 200). 3. Wait for the next retry processor cycle. 4. Check: (a) RabbitMQ paused queue message count decreased, (b) WORKING endpoint received the previously-failed callbacks.	• Paused queue drains: message count in RabbitMQ decreases (goes to 0 if all retried). • WORKING endpoint receives the retried callbacks. • Events.Service logs confirm redelivery and success acknowledgment.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-RETRY-003 <i>Status transition: Paused → Active on successful re-delivery</i>	P1-Critical QA Status: ☐ Open	- Continue from TC-RETRY-002 (subscription fixed, messages retried). - After the retry cycle completes, call GET /api/v1/study/{id}/subscriptions. - Check the 'status' field of the previously-paused subscription.	- Subscription status is now 'Active' (not Paused, not Stopped). - UI Subscription Configuration tab reflects Active status for this subscription.
TC-RETRY-004 <i>Status transition: Paused → Stopped + email after threshold exceeded</i>	P1-Critical QA Status: ☐ Open	- Pre-condition: Keep the FAILING endpoint in place (do NOT fix it). Allow multiple retry cycles to pass. - Monitor: each retry cycle should fail again. After N failures (N = configured threshold), the subscription should move to Stopped. - Watch the test mailbox during the cycles. - After Stopped state is reached, call GET /api/v1/study/{id}/subscriptions.	- After threshold failures, subscription status = 'Stopped' (not Paused, not Active). - A 'subscription stopped' email notification is sent to the test mailbox with correct subscription details. - No further delivery attempts are made once status is Stopped (confirm via WORKING endpoint — no new POSTs).

3.8 Health Checks, Swagger & OIDC (TC-HEALTH-*)

Risk area: SqlClient upgrade could cause health check dependency failures. Swagger root redirect removal is a deliberate change — confirm it is removed but Swagger still works at /swagger.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-HEALTH-001 <i>Health endpoints – all 5 services healthy under normal conditions</i>	P1-Critical QA · STG · PROD Status: ☐ Open	- For each of the 5 integration services, call the health endpoint (typically GET /health or GET /healthz). - For services with SQL dependency: confirm the SQL check is included in health response. - For services with Redis dependency (if applicable): confirm Redis check included.	- HTTP 200 from each health endpoint. - Response body shows status: 'Healthy' (or equivalent). - SQL connectivity check passes (no SqlClient regression from Microsoft.Data.SqlClient upgrade). - No 'Degraded' or 'Unhealthy' status under normal load.
TC-HEALTH-002 <i>Swagger UI – accessible only when EnableSwagger=true, root redirect REMOVED</i>	P1-Critical QA · STG Status: ☐ Open	- Confirm EnableSwagger is true in the QA environment configuration. - Navigate to /swagger/index.html for each API service — verify Swagger UI loads. - Navigate to / (root path) of each API service — verify it does NOT redirect to /swagger/index.html. - In a Stage/Prod environment where EnableSwagger is false: confirm /swagger/index.html returns 404 (not accessible).	- Swagger UI loads at /swagger/index.html when EnableSwagger=true. - Root path / does NOT redirect to swagger (the redirect was removed — accessing / should return 404 or a minimal response). - When EnableSwagger=false: /swagger/index.html returns 404 (Swagger is not exposed).

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-HEALTH-003 <i>OIDC login flow – redirects and query parsing work correctly</i>	P2-High QA Status: ☐ Open	1. If OIDC login flow is used in this environment: initiate a fresh login session by clearing cookies. 2. Complete the OIDC login flow end-to-end. 3. After login, confirm redirect back to the application lands on the correct page (not a blank page or error). 4. In logs, confirm MedrioIdpConnectHandler processed the callback correctly (no query parsing errors).	• OIDC login completes without error. • Post-login redirect lands on the expected application page. • No auth loop (not redirected back to login again after completing login). • No 'invalid_state' or 'correlation failed' errors in logs.

3.9 Backward Compatibility (TC-BACK-*)

Risk area: Pre-existing customer studies must work without any reconfiguration. The upgrade must be transparent to all configured studies.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-BACK-001 <i>Existing configured study – no reconfiguration needed</i>	P1-Critical QA · STG · PROD Status: ☐ Open	1. Identify a study that was configured BEFORE this deployment (pre-existing CTMS mapping + active subscriptions). 2. WITHOUT making any changes, load: (a) Study Detail page (CTMS mapping visible), (b) Subscription Configuration tab (subscriptions listed), (c) Click 'Go to CTMS' (redirect works). 3. Trigger an event on this pre-existing study and confirm delivery to its subscriber.	• All three touchpoints work without any reconfiguration by the QA engineer. • CTMS mapping displays the pre-existing value (not cleared by the upgrade). • Subscriptions are listed (not deleted by the upgrade). • Event delivery succeeds to the pre-existing subscriber endpoint.
TC-BACK-002 <i>Plain study (CTMS OFF, Events OFF) – no unexpected integrations calls</i>	P2-High QA Status: ☐ Open	1. Open Study D (CTMS OFF, Events OFF) in Admin Portal. 2. Load Study Detail page. Open DevTools → Network tab. 3. Observe: are any integrations API calls made (to /api/integrations/... or /api/v1/...)? 4. Perform a form submission or import on Study D. 5. Check Events.Service logs — any event generated for Study D?	• Study Detail page loads with NO outbound call to any integration service. • No CTMS field visible in the UI for Study D. • No event generated or delivered for Study D actions. • No error in integration service logs referencing Study D.

3.10 Non-Functional Smoke (TC-PERF-*)

Risk area: .NET 10 / EF Core 10 runtime changes can affect memory and throughput. These are smoke-level checks — not full load tests.

TC ID	Env / Priority	Test Steps (Pre-condition → Action → Verification)	Expected Result (Pass Criteria)
TC-PERF-001 <i>API latency stability – no progressive slowdown over 30 calls</i>	P2-High QA Status: ☐ Open	- Using Postman Collection Runner or a simple script, call GET /api/v1/study/{studyId}/subscriptions 30 times in sequence (1 call per second). - Record response time for each call. - Plot or compare first 5 calls vs last 5 calls response times.	- Response times remain stable (within $\pm 20\%$ variance across all 30 calls). - No progressive increase (e.g., call 1=120ms, call 30=800ms would be a defect — suggests connection pool exhaustion or memory pressure). - No timeouts or errors within the 30 calls.
TC-PERF-002 <i>CPU/Memory snapshot – no abnormal spike during event processing</i>	P2-High QA Status: ☐ Open	- While TC-EVT-001 through TC-EVT-004 are running (event processing active), open the container/process monitor for all 5 integration services. - Record CPU % and memory (RSS/Heap) for each service during processing. - Compare against baseline (if available from the previous .NET 6 build).	- CPU does not spike above 80% sustained during normal event processing. - Memory usage does not grow unboundedly across successive events (no memory leak). - All 5 services remain stable — no OOM kill or restart during the test window.

4. Environment Execution Matrix

The table below defines exactly which test cases to execute per environment. ☒ = execute, **⏏** = intentionally skip (reason in Notes).

Test Case Group	FB	Master	Stage	Notes
TC-AUTH-001 Valid token	☒ Full	☒ Full	☒ Smoke	Run in PROD as first smoke test after deployment
TC-AUTH-002 Invalid token	☒ Full	⏏ Skip	⏏ Skip	Security negative test — QA env only
TC-AUTH-003 IDP timeout	☒ Full	⏏ Skip	⏏ Skip	Requires proxy/Toxiproxy — QA only
TC-AUTH-004 Header injection	☒ Full	☒ Verify logs	⏏ Skip	Log trace available in QA + STG
TC-SUBS-001 View list	☒ Full	☒ Full	☒ Smoke	Read-only — safe in all envs
TC-SUBS-002 Create	☒ Full	☒ Full	⏏ Skip	Do NOT create test data in PROD
TC-SUBS-003 Edit	☒ Full	☒ Full	⏏ Skip	No data mutations in PROD
TC-SUBS-004 Delete	☒ Full	☒ Full	⏏ Skip	No data mutations in PROD
TC-SUBS-005 Empty state	☒ Full	⏏ Skip	⏏ Skip	Requires Study B setup
TC-SUBS-006 Failing endpoint	☒ Full	⏏ Skip	⏏ Skip	Generates errors — QA only

Test Case Group	FB	Master	Stage	Notes
TC-CTMS-001 Mapping load	☑ Full	☑ Full	☑ Smoke	Read-only — safe in all envs
TC-CTMS-002 Set mapping	☑ Full	☑ Full	⏸ Skip	No data mutations in PROD
TC-CTMS-003 Clear mapping	☑ Full	⏸ Skip	⏸ Skip	Potentially destructive
TC-CTMS-004 No mapping	☑ Full	⏸ Skip	⏸ Skip	Requires Study C setup
TC-REDIR-001 Redirect happy path	☑ Full	☑ Full	☑ Smoke	Read-only redirect — safe in PROD
TC-REDIR-002 No mapping safe fail	☑ Full	⏸ Skip	⏸ Skip	Requires Study C
TC-REDIR-003 Security (unauth)	☑ Full	⏸ Skip	⏸ Skip	Security negative — QA only
TC-EVT-001 ODM import events	☑ Full	☑ Full	⏸ Skip	Import triggers may affect PROD data
TC-EVT-002 Excel import events	☑ Full	⏸ Skip	⏸ Skip	QA only
TC-EVT-003 Data entry events	☑ Full	⏸ Skip	⏸ Skip	QA only
TC-EVT-004 RecalcVariables events	☑ Full	⏸ Skip	⏸ Skip	QA only
TC-EVT-005 No subscribers	☑ Full	⏸ Skip	⏸ Skip	Requires Study B
TC-EVTAPI-001 API smoke	☑ Full	☑ Full	⏸ Skip	Direct API call — verify connectivity
TC-EVTAPI-002 Failure email	☑ Full	⏸ Skip	⏸ Skip	Requires failing endpoint setup
TC-RETRY-001 Processor fires	☑ Full	⏸ Skip	⏸ Skip	Requires paused state
TC-RETRY-002 Queue drain	☑ Full	⏸ Skip	⏸ Skip	Requires paused queue
TC-RETRY-003 Paused→Active	☑ Full	⏸ Skip	⏸ Skip	State machine validation
TC-RETRY-004 Paused→Stopped email	☑ Full	⏸ Skip	⏸ Skip	State machine + email validation
TC-HEALTH-001 Health endpoints	☑ Full	☑ Full	☑ Smoke	Immediate PROD check post-deploy
TC-HEALTH-002 Swagger behavior	☑ Full	☑ Full	⏸ Skip	Swagger likely disabled in PROD
TC-HEALTH-003 OIDC flow	☑ Full	⏸ Skip	⏸ Skip	If OIDC applicable in env

Test Case Group	FB	Master	Stage	Notes
TC-BACK-001 Pre-existing study	☑ Full	☑ Full	☑ Smoke	Critical backward compat — run in PROD
TC-BACK-002 Plain study	☑ Full	⏭ Skip	⏭ Skip	QA setup required
TC-PERF-001 Latency stability	☑ Full	⏭ Skip	⏭ Skip	Script-based — QA only
TC-PERF-002 CPU/Memory snapshot	☑ Full	⏭ Skip	⏭ Skip	Container monitoring — QA only

5. Production Go-Live Smoke Checklist

Run IMMEDIATELY after deploying to Production. If ANY item fails, initiate rollback. These are read-only/non-destructive checks.

#	Smoke Check	Pass Criterion / Rollback Trigger
1	GET /health on all 5 integration services Status: ☐ Open	HTTP 200 + 'Healthy'. ANY non-200 = ROLLBACK.
2	Load Subscription Configuration tab for a pre-existing configured study Status: ☐ Open	Subscription list renders with no JS errors. HTTP 200 from backend.
3	Load Study Detail page → CTMS Customer ID displays Status: ☐ Open	Pre-existing mapping value visible. No empty field where value should be.
4	Click 'Go to CTMS' for a pre-existing mapped study Status: ☐ Open	Redirect succeeds. Final CTMS URL correct. No 401/500.
5	Trigger one low-risk event (e.g., open/close a form) on a pre-existing study with an active subscription Status: ☐ Open	Event delivered to subscriber within 60s (confirm with subscriber team). No error in Events.Service log.
6	Confirm Swagger is NOT accessible at root / Status: ☐ Open	Root / returns 404 or plain response — NOT a Swagger redirect. (Security: Swagger should not be publicly exposed in PROD.)
7	Call GET /api/v1/study/{knownStudyId}/subscriptions with a valid token Status: ☐ Open	HTTP 200 with expected subscription list. No auth regression.

NOTE: If all 7 checks pass: deployment is confirmed good. Notify the team and close the release checklist.

6. Pass / Fail Criteria & Defect Prevention

6.1 Overall Pass Criteria (ALL must be true)

	Criterion	Covers TCs
✓	All P1-Critical test cases pass in QA with zero defects.	TC-AUTH-001/002/003, TC-SUBS-001/002/006, TC-CTMS-001/002, TC-REDIR-001/002/003, TC-EVT-001/002/003, TC-EVTAPI-001/002, TC-RETRY-001-004, TC-HEALTH-001/002, TC-BACK-001
✓	All P2-High test cases pass or are accepted with documented justification.	TC-AUTH-004, TC-SUBS-003/004/005, TC-CTMS-003/004, TC-EVT-004/005, TC-HEALTH-003, TC-BACK-002, TC-PERF-001/002
✓	Zero HTTP 500 responses across all API calls during the entire test execution.	All TCs
✓	Zero stack traces or internal exception details exposed in API response bodies.	All TCs
✓	Auth failures always return 401 or 403 — never 200, never 500.	TC-AUTH-002, TC-REDIR-003
✓	Subscription status machine transitions correctly: Active → Paused → Active/Stopped.	TC-RETRY-003, TC-RETRY-004
✓	Email notifications are sent with correct content for both failure and recovery events.	TC-EVTAPI-002, TC-RETRY-004
✓	All 7 Production smoke checklist items pass after deployment.	Section 5

6.2 Defect Prevention — Known Upgrade Risk Areas

These are the most likely defect hiding spots given the specific changes in this upgrade. Double-check these during execution:

Risk Area	What to Watch For	Where to Look
OAuth2 introspection misconfiguration	All EDC→Integrations calls silently returning 401/403. No obvious error in UI — just empty lists or non-functional buttons.	<i>Integration service startup logs: look for 'Authority', 'IntrospectionEndpoint', 'ClientId' being read correctly.</i>
Medrio-UserType header missing	IDP rejects introspection silently, causing auth failures that look like network issues.	<i>Outbound HTTP trace on integration service → IDP backchannel call must include this header.</i>
EF Core query behavioral change	CTMS mapping or subscription data returning null/empty when data exists in DB. OR: extra DB roundtrips causing latency.	<i>SQL Server profiler or EF Core logs (set log level to Debug for EF) to see generated SQL queries.</i>
NServiceBus message format change	Paused queue messages from BEFORE the upgrade cannot be deserialized by the new NServiceBus version. Silent message loss.	<i>Test by checking if any pre-existing paused messages (before deploy) can still be processed after deploy.</i>
Serilog sink configuration change	Log output format changes causing monitoring/alerting to miss log entries. Or PII unexpectedly logged.	<i>Check log output format in QA matches expected JSON schema. Grep for email addresses, patient IDs in logs.</i>

Risk Area	What to Watch For	Where to Look
Docker cert store difference	HTTPS calls from integration service to IDP or subscriber endpoints failing due to missing root CA in .NET 10 base image.	Look for SSL/TLS handshake errors in integration service logs when calling external HTTPS endpoints.
.NET 10 HTTP client defaults	Default timeout, connection pool, or redirect behavior changed — could affect subscriber delivery or IDP calls.	Observe TC-AUTH-003 and TC-EVT-* — any call that suddenly times out faster or slower than before.
Swagger root redirect removal	If the root redirect was NOT removed, a client navigating to / could be unintentionally exposed to Swagger in environments where EnableSwagger=true.	Explicitly test: GET / should NOT return a 302 to /swagger in any environment.

7. Defect Triage Priority Guidelines

7.1 Defect Triage Priority Guide

Severity	Condition	Action
P1 – Blocker	Any HTTP 500, any auth bypass, any data loss, Production smoke fails	STOP deployment. Rollback if in PROD. Immediate dev fix required before re-testing.
P2 – Critical	TC-AUTH / TC-REDIR / TC-RETRY failure, email not sent, subscription stuck in wrong state	Block release. Fix and re-run affected TC section before proceeding.
P3 – Major	Swagger exposed unintentionally, OIDC minor glitch, latency regression >50%	Document and fix before next release. Do not ship knowingly.
P4 – Minor	UI cosmetic, log format difference, minor wording	Acceptable to defer with documented exception. Log as tech debt.